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#include <Servo.h>

int ledRed = A0;
int ledYellow = A1;
int ledBlue = A2;

Servo
servo1;
Servo servo2;
Servo servo3;
Servo servo4;
Servo servo5;

int pinServo1 = 9;
int
pinServo2 = 10;
int pinServo3 = 11;
int pinServo4 = 12;
int pinServo5 = 13;

int
posStartServo1 = 50;
int posEndServo1 = 90;

int posStartServo2 = 25;
int posEndServo2 =
0;

int posStartServo3 = 70;
int posEndServo3 = 120;

#define S0 4
#define S1 5
#define
S2 6
#define S3 7
#define sensorOut 8
int frequencyRed = 0;
int frequencyGreen = 0;
int
frequencyBlue = 0;

bool colorFind = false;

void setup() {
    pinMode(ledRed, OUTPUT);

    pinMode(ledYellow, OUTPUT);
    pinMode(ledBlue, OUTPUT);

    servo1.attach(pinServo1);

servo2.attach(pinServo2);
    servo3.attach(pinServo3);
    servo4.attach(pinServo4);

servo5.attach(pinServo5);

    moveServoStart();

    pinMode(S0, OUTPUT);
    pinMode(S1,
OUTPUT);
    pinMode(S2, OUTPUT);
    pinMode(S3, OUTPUT);
    pinMode(sensorOut, INPUT);

digitalWrite(S0, HIGH);
    digitalWrite(S1, LOW);

    Serial.begin(9600);
}
void loop() {

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getFrequency();

    if (!colorFind) {
        checkColor();
    } else {
        moveBlock();
    }
}

void getFrequency() {
    digitalWrite(S2, LOW);
    digitalWrite(S3, LOW);

    frequencyRed = pulseIn(sensorOut, LOW);
    Serial.print("Rouge = ");

    Serial.print(frequencyRed);
    Serial.print(" ");

    delay(100);

    digitalWrite(S2, HIGH);
    digitalWrite(S3, HIGH);
    frequencyGreen = pulseIn(sensorOut,
LOW);
    Serial.print("Verte= ");
    Serial.print(frequencyGreen);

    Serial.print(" ");

    delay(100);
    digitalWrite(S2, LOW);

    digitalWrite(S3, HIGH);
    frequencyBlue = pulseIn(sensorOut, LOW);
    Serial.print("Bleu=
");
    Serial.print(frequencyBlue);
    Serial.println(" ");

    delay(100);
}

void checkColor() {
    digitalWrite(ledRed, LOW);
    digitalWrite(ledYellow,
LOW);
    digitalWrite(ledBlue, LOW);
    if ((frequencyRed < 45 && frequencyRed >
25)
        && (frequencyGreen < 105 && frequencyGreen > 85)
        && (frequencyBlue < 75 && frequencyBlue > 55)) {
        digitalWrite(ledRed, HIGH);
        colorFind = true;
    } else if ((frequencyRed < 45
&& frequencyRed > 25)
        && (frequencyGreen < 60 &&
frequencyGreen > 40)
        && (frequencyBlue < 85 &&
frequencyBlue > 65)) {
        digitalWrite(ledYellow, HIGH);
        colorFind = true;
    } else
    if ((frequencyRed < 90 && frequencyRed > 70)
        &&
        (frequencyGreen < 95 && frequencyGreen > 75)
        &&
        (frequencyBlue < 60 && frequencyBlue > 40)) {
        digitalWrite(ledBlue, HIGH);

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    colorFind = true;
}
}

void moveServoStart() {
    servo1.write(posStartServo1);

    servo2.write(posStartServo2);
    servo3.write(posStartServo3);
    servo4.write(90);

    servo5.write(90);
}

void moveBlock() {
    clawClose();

    setServoPosition(servo2,
posStartServo2, posEndServo2);
    setServoPosition(servo3, posStartServo3, posEndServo3);

    setServoPosition(servo1, posStartServo1, posEndServo1);

    setServoPosition(servo1,
posEndServo1, posStartServo1);
    setServoPosition(servo3, posEndServo3, posStartServo3);

    setServoPosition(servo2, posEndServo2, posStartServo2);

    delay(2000);

    clawOpen();

    moveServoStart();

    colorFind = false;
}

void setServoPosition(Servo servo, int
startPosition, int endPosition) {
    if (startPosition > endPosition) {
        for (int i =
startPosition; i >= endPosition; i--) {
            servo.write(i);
            delay(30);
        }
    }
    else {
        for (int i = startPosition; i <= endPosition; i++) {
            servo.write(i);

            delay(30);
        }
    }
}

void clawOpen() {
    servo4.write(70);

    servo5.write(100);
}

void clawClose() {
    servo4.write(100);
    servo5.write(80);
}

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